

ScribeNova – A SaaS Platform for AI-Powered Content Generation and Interactive Document Querying

A Synopsis submitted in partial fulfilment of the requirement for the award of
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1. Introduction:

The increase in content sites such as YouTube, LinkedIn and Instagram have ensured that there are heightened expectations for continuous, relevant and standard material. Researchers and practitioners tend to work with voluminous texts, for instance research papers, manuals and reports, that are also quite complex to reference without extra efforts. In response to these calls, ScribeNova is put forth as a complete SaaS web platform which provides a solution. The platform allows the creation of content that can be optimized to each platform, and also allows users to interact with PDF documents using ordinary language. With its powerful abilities, ScribeNova extracts vital information from the document, answers, accompanied by metadata elements which include, page number, line number and PDF filename.

ScribeNova uses sleek and easy interfaces powered by cutting-edge technology such as generative AI and RAG. Through investigating this initiative, we demonstrate the capability of AI to solve everyday issues and demonstrate a pioneer approach that could enhance both creativity and productivity. It differs from a regular tool, serving more like a multifunction digital assistant meant for use by creators, professionals, and students.

2. Problem Statement:

In today's fast-paced digital environment, users face two major productivity bottlenecks: the lack of streamlined tools for high-quality content creation and inefficient methods of extracting information from long-form documents.

- **Content Creation Challenges:** As for digital marketers, influencers, and professionals, there is one frequently difficult task: creating content specific to given platforms, which keeps people interested and ensures the consistency of content. This need for switching to various tools and doing manual edits results in a long workflow for many users.
- **Inefficient Document Navigation:** Scientists, lawyers as well as students of law often come across long PDF files when doing their work. A traditional keyword search frequently omits contextual details, which leads to poorer and slower information search results.
- **Fragmented Tool Ecosystem:** Modern solutions focus on content generation or analysis of documents, as it were, instead of supporting both processes simultaneously. Such lack of consolidation makes it cumbersome to navigate through apps and areas of fluidity and effectiveness of the workflow are compromised.

ScribeNova addresses these problems by merging content production and document manipulation on a single AI-empowered SaaS platform that creates an efficient and unified workflow.

3. Objective:

The key goal of ScribeNova is to present a state-of-the-art, AI-driven SaaS service that will allow users to conduct content production and document management more effectively. It aims to give users the power to create a platform that is intuitive for building content that is tailor-made for channels like YouTube, Instagram, LinkedIn, and many other digital contexts. In addition, the project incorporates advanced AI features that empower the users to interrogate PDF documents in natural language, guaranteeing that information extraction is done with maximum precision and speed.

ScribeNova is building on advanced frameworks and APIs, including Gemini for content creation and LangChain for data retrieval to produce the desired output. More so, secure login is done through Clerk authentication while it uses scalable databases such as Neon and Drizzle ORM for best functions. Our goal is to build a system oriented toward user needs that can give responsiveness, scalability and accuracy in real creative and professional situations.

4. Problem Solution:

To address the issues discussed in the problem statement, ScribeNova proposes a fully integrated, intelligent solution, which combines the AI-generated content with highly advanced capabilities of document interaction.

- **Content Generation Module:** With the integration of Gemini API, the module produces platform-specific content for YouTube, LinkedIn, Instagram, reworked articles and e-commerce descriptions. It simplifies creative processes, allowing users the ability to do more with resources on meaningful decisions by not getting caught up in trivial tasks.
- **AI Chat with PDF Module:** Users can upload a PDF and chat with the system in ordinary terms. Using LangChain and Qdrant, the platform finds relevant content portions and provide answers that include such details as a PDF's title, the page number it is found on, and the exact line of that page. This enhances the task of research and reviewing of documents for the users.
- **Integrated Dashboard:** Both the modules are available with a common interface compatible for convenience. With the intuitive design, users can easily switch between different functions and the problem is worked out for the both technical experts and general users.

The flexible and amenable structure of ScribeNova makes it possible for the application to adapt in order to accommodate increasing user needs by adding more content formats and document types to future releases.

5. System Architecture and Technologies Used:

ScribeNova employs a layered modular architecture designed to provide the application with the ability to scale, offer security as well as provide optimal performance. Each element of the system is marked with a unique function, which promotes modular development and a simple maintenance. Based on principles of modern microservices, the architecture uses stateless APIs, and distributed storage for data.

Frontend Layer: Converting to Next.js, the platform allows fast loading of content with better SEO by having server-side rendering capabilities and SSG capabilities. The user interface is built with device responsiveness in mind, thereby, easily accessible and user-friendly to both desktop and mobile users.

Backend API Layer: Developed using the Node.js, this component takes care of orchestration of business logic, API request, and the integration with different third-party platforms. It promotes content creation as well as interaction with materials using crafted server requests.

AI and RAG Layer: ScribeNova's fundamental intelligence is a combination of:

- Gemini API is charged with the task of producing high quality articles.
- The LangChain is used to build chains where the language models may communicate with structured and unstructured data.
- Ollama is a method to quickly handle locally processed language models in a low resource friendly way.
- To enable semantic searches and find similarities of documents stored in PDF files, the Qdrant, a vector database, is implemented.
- Valkey acts as a good cache layer and reduces the time taken to make API requests.

Database Layer: Harnesses Drizzle ORM from the Neon (PostgreSQL) platform to aggregate structured data such as user profiles, sessions, and contents used. The relational structure has been shaped to ensure the integrity of data and to facilitate scalable operations.

Authentication Layer: This layer, maintained by Clerk, is of great strength in terms of user management security, which includes, sign up, sign in, session handling, and access control using JWT and RBAC.

Technology Stack Overview:

Component	Technology Used
Frontend	Next.js (React-based framework for UI & SSR)
Backend	Node.js (Event-driven runtime for APIs)
AI Integration	Gemini API, LangChain, Ollama
Vector Search Engine	Qdrant (for semantic similarity search)
Authentication	Clerk (Secure user login, session management)
Database	Drizzle ORM + Neon PostgreSQL (Structured data persistence)
Caching	Valkey (High-speed in-memory caching)
PDF Parsing	LangChain Document Loaders

6. Advantages:

All-in-One Functionality:

Combines content creation and PDF interaction into a single platform, eliminating the need to switch between tools.

Platform-Specific Content Generation:

Supports optimized content output for YouTube, Instagram, LinkedIn, and more, enhancing user engagement and reach.

AI-Enhanced Productivity:

Uses advanced LLMs and RAG to automate repetitive and time-consuming tasks like writing, paraphrasing, and document searching.

Metadata-Rich Responses:

PDF queries return contextual results with page number, line number, and file name, improving traceability and research efficiency.

Secure and Scalable:

Employs robust authentication (Clerk), scalable databases (Neon), and efficient data handling practices to ensure system integrity.

Modern UI/UX Design:

Intuitive interface built with Next.js and React provides a smooth and responsive user experience on all devices.

Extensibility:

Designed with modular architecture, allowing new features, formats, or integrations to be added with minimal refactoring.

7. Future Scope

ScribeNova has the potential to evolve into a comprehensive AI productivity ecosystem. The following are the future enhancements envisioned for the platform:

Multimodal Content Support

- **Audio/Video Processing:** Enable users to upload videos or audio files for summarization, transcription, or content repurposing.
- **Visual Input Handling:** Use OCR and image recognition to process scanned documents, infographics, or handwritten notes.

Real-Time Collaboration Tools

- **Team Workspaces:** Introduce shared project boards for group-based content creation and research.
- **Live Editing and Annotation:** Allow multiple users to chat with the same PDF or co-write content with version tracking.

Learning Management System (LMS) Integration

- **Course Material Generation:** Automatically summarize textbooks, notes, and research papers into lectures or quizzes.
- **Assignment Helpers:** Support students with AI-powered document analysis and citation generation.

Advanced Analytics Dashboard

- **Content Performance Tracking:** Provide engagement metrics for generated content across platforms (likes, shares, views).
- **Query Insights:** Show trends, frequent topics, and question accuracy for PDF interactions.

Plugin and Integration Marketplace

- **Custom AI Agents:** Let users build or install specialized AI agents (e.g., resume writer, legal brief summarizer).
- **Tool Integration:** Seamlessly connect with tools like Notion, Google Docs, Slack, or Trello.

Smart Template System

- **Auto-Adaptive Templates:** Generate layouts and formats based on user goals (e.g., blog post, report, Instagram reel).
- **User-Saved Presets:** Allow users to create and reuse custom templates across different projects.

8. References:

Frontend References

1. **Next.js Documentation**
 - <https://nextjs.org/docs>
 - Official guide to building server-rendered React apps with routing, SSR, SSG, and API capabilities.

2. **ShadCN UI Component Library**
 - o <https://ui.shadcn.dev>
 - A modern UI library based on Tailwind CSS used for building beautiful and accessible interfaces.
3. **React PDF Viewer**
 - o GitHub Repository: [react-pdf-viewer](https://github.com/wojtekmaj/react-pdf-viewer)
 - Open-source PDF viewer library used for rendering and interacting with PDF documents in React.

Backend References

1. **Node.js Documentation**
 - o <https://nodejs.org/en/docs>
 - Core documentation for building scalable server-side applications using Node.js.
2. **Express.js Web Framework**
 - o <https://expressjs.com>
 - Fast, unopinionated, minimalist web framework for Node.js used to handle API routes and middleware.
3. **LangChain for Node.js**
 - o <https://js.langchain.com>
 - Toolkit for building context-aware LLM apps, supporting PDF loading, prompt chaining, and retrieval-based QA.

AI & RAG References

1. **Gemini API by Google**
 - o <https://ai.google.dev>
 - Google's large language model API used for content generation and smart assistance.
2. **Ollama (Local LLM Runtime)**
 - o <https://ollama.com>
 - Tool to run open-source LLMs like LLaMA, Mistral, etc., locally for offline-friendly AI processing.
3. **Qdrant Vector Search Engine**
 - o <https://qdrant.tech>
 - Open-source vector similarity search engine used for efficient retrieval in RAG pipelines.

Database References

1. **Neon PostgreSQL**
 - o <https://neon.tech>
 - Serverless PostgreSQL database designed for modern cloud-native applications with branching and autoscaling.
2. **Drizzle ORM**
 - o <https://orm.drizzle.team>
 - Type-safe SQL ORM for JavaScript/TypeScript used for schema migrations and query building.
3. **Valkey (Redis Alternative)**
 - o <https://valkey.io>

- High-performance in-memory caching engine, forked from Redis, used for optimizing job queues and session storage.

Authentication References

1. **Clerk Authentication**
 - o <https://clerk.dev/docs>
 - Plug-and-play user authentication platform offering sign-up/sign-in, JWTs, and RBAC for modern apps.
2. **JWT (JSON Web Tokens)**
 - o <https://jwt.io/introduction>
 - Secure and compact method for transmitting information between parties, often used for session validation.
3. **RBAC (Role-Based Access Control)**
 - o GitHub Repository: [RBAC Example in Node.js](#)
 - Practical examples of implementing access control based on user roles and permissions.

9. Timeline:

Task Name	FEBRUARY	MARCH	APRIL	MAY
Selection of Project Name	Finalized "ScribeNova"			
Synopsis Submission	Drafted and submitted			
Research & Planning	Feature Planning, Tech Stack Selection	Finalized Architecture		
Frontend Development		UI Mockups and Component Planning	Next.js and Tailwind UI Development	Ongoing: Integration with Backend APIs
Backend Development		API Skeleton Setup	Node.js API, LangChain, Qdrant Setup	Finalizing Clerk Auth, Valkey, Drizzle ORM
AI & PDF Module Implementation		Dataset Curation	PDF Chat using LangChain and Metadata	Fine-tuning Gemini Content Generation
System Integration & Dashboard			Started Integration Work	Finalizing Unified Dashboard
Testing & Debugging				Starting this week

Deployment Preparation				Preparing for Vercel Deployment
Final Report Submission				Final Report and Documentation